

CodeAlpha Internship

Task 2: Automatic Street Light Using LDR and Arduino

Submitted By

Name: Marwa Bibi

Intern ID: CA/DF1/176025

University: University of Engineering and Technology, Taxila

Department: Telecommunication Engineering

Date: 10 July 2026

1. Objective

To design an automatic street light system using Arduino Uno and an LDR. The LED turns ON in darkness and OFF in bright light.

2. Components Required

- Arduino Uno R3
- Breadboard
- LDR
- LED
- 220 Ω , 10 k Ω Resistor
- Jumper Wires
- Tinkercad

3. Circuit Description

The LDR and the 10 k Ω resistor form a voltage divider. The voltage at their junction is connected to the Arduino analog pin A0. The Arduino continuously reads the light intensity. The LED is connected to digital pin 13 through a 220 Ω resistor. Depending on the light intensity, the Arduino turns the LED ON or OFF.

4. Working Principle

The Arduino continuously reads the analog value from the LDR through pin A0. If the light intensity is below the threshold value, the Arduino turns the LED ON. Otherwise, it turns the LED OFF.

5. Arduino Program

The Arduino program reads the LDR value from analog pin A0 and controls the LED connected to digital pin 13. A screenshot of the program is shown below.

```

1  int ldr = A0;
2  int led = 13;
3
4  void setup() {
5      pinMode(led, OUTPUT);
6      Serial.begin(9600);
7  }
8
9  void loop() {
10     int value = analogRead(ldr);
11     Serial.println(value);
12
13     if (value < 500) {
14         digitalWrite(led, HIGH);
15     } else {
16         digitalWrite(led, LOW);
17     }
18
19     delay(200);
20 }

```

Figure 1: Arduino Program

6. Circuit Diagram

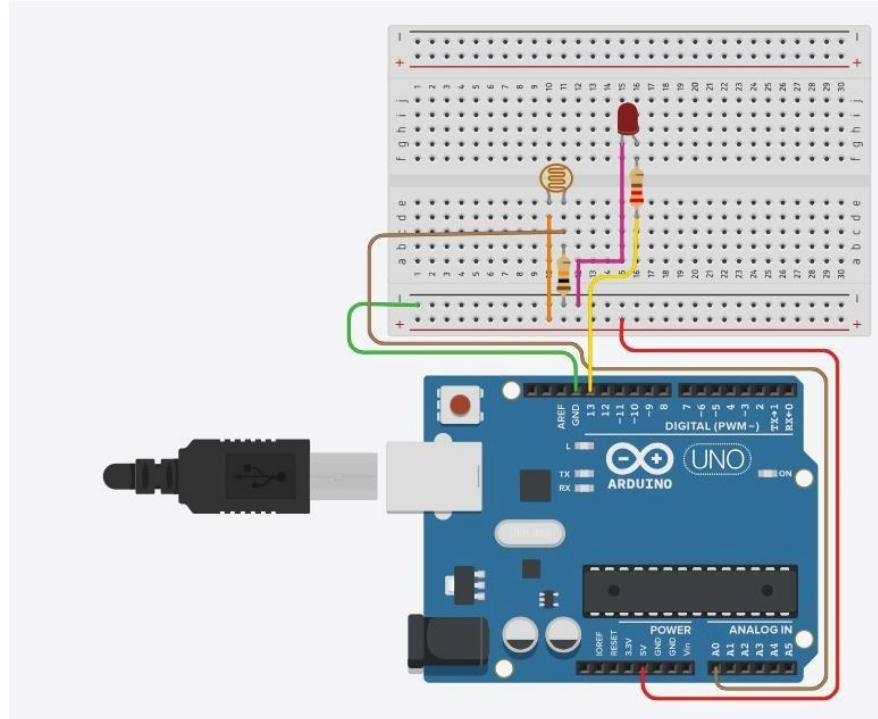


Figure 2: Complete Circuit Diagram

7. Simulation Results

- **Dark Condition**

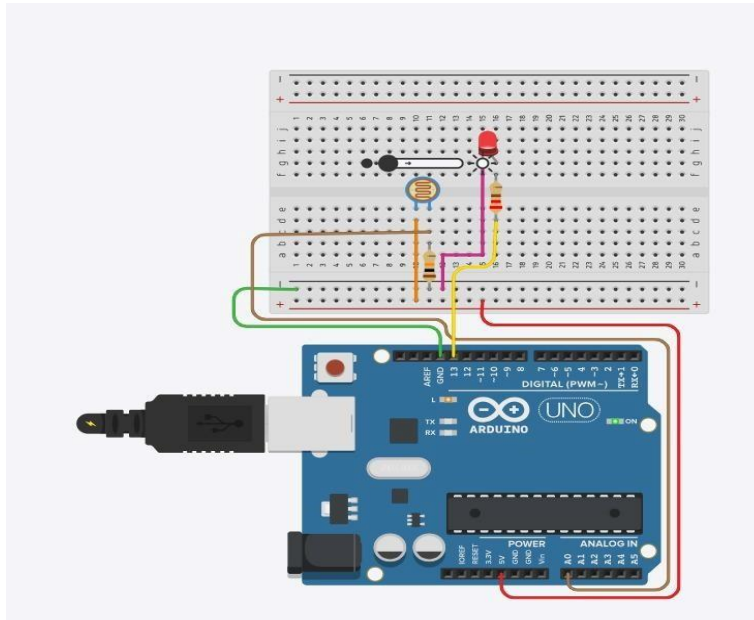


Figure 3: LED ON in Dark Condition

- **Bright Condition**

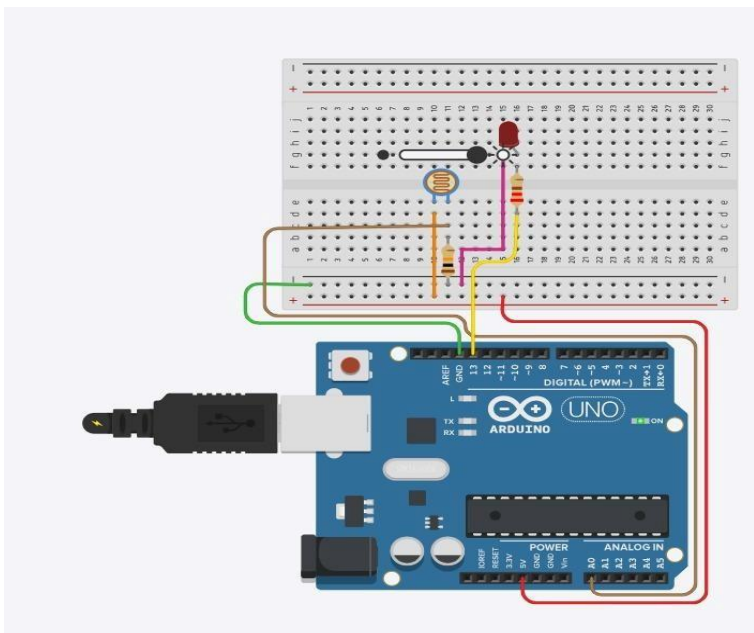


Figure 4: LED OFF in Bright Condition

8. Applications

- Automatic street lights
- Home automation
- Garden lighting
- Energy saving systems

9. Advantages

- Low cost
- Automatic operation
- Reliable
- Energy efficient

10. Conclusion

The project successfully demonstrates automatic lighting using an LDR and Arduino.